

SYNTHESIS OF IMIDAZOLINE COMPLEXES OF PETROLEUM ACIDS, RESEARCH AND APPLICATION OF THEIR PETROLEUM-COLLECTING AND DISPERSING, ELECTRICAL CONDUCTIVITY PROPERTIES

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Abstract

This article presents techniques for safeguarding water bodies in the event of oil spills at underwater crossings of both stationary and mobile oil pipelines. Additionally, a device for collecting oil beneath the surface of ice is introduced. Their technical description and operational principle are provided. Research has confirmed that the oil-holding capacity of the sorbent remains constant regardless of the length of time it spends in the water. The synthesized product was found to have high surface activity at the water-air interface, as determined by tensiometric measurements. Additionally, its colloidal-chemical parameters were calculated. Lab tests have demonstrated that both the reagent itself and its 5% aqueous solution possess a significant ability to collect oil.

Keywords: oil collection, petroleum acid amidoamines, amino ester, surface activity.

Currently, a major global issue is the mitigation of water pollution and the remediation of contaminated water. The dispersion of oil during production and transportation can contaminate soil and water, resulting in pollution and an imbalance in the ecological equilibrium between the atmosphere and the hydrosphere. It is evident that while a thick layer of oil can be cleaned using mechanical methods, it is not feasible to remove a thin layer of oil through mechanical means. Surfactants are employed to cleanse the fragile oil film, rendering them ecologically benign and capable of efficiently interacting with oil and petroleum substances. The chloride salt is obtained from the chloric acid of amino acids synthesized using gastrointestinal methylatanolamines. Research has been conducted on the characteristics of the surface activity of amino acids and their modifications. The oil and gas potential of the salt that was extracted was investigated by examining a thin layer of oil spill (~0.17 mm) on the surface of three different types of mineral water (sea, drinking, and distilled).

Currently, there is a growing body of scientific research focused on exploring the appeal of natural raw materials for different chemical processes. These raw materials can be reclaimed and are ecologically sustainable. Hence, it is crucial and pertinent to develop and investigate novel efficient surfactants derived from natural sources [1-3].

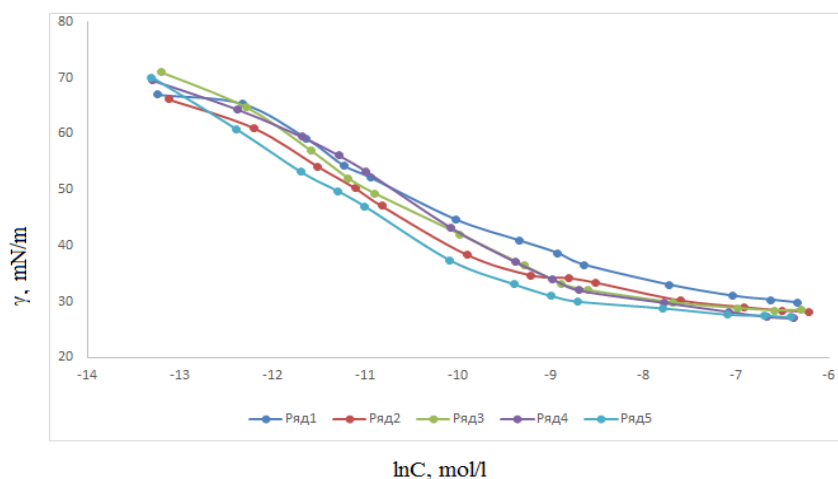
This article presents the results of study of synthesis of chlorine-containing amidoamine and imidazoline complexes of petroleum acids. The initial compounds were synthesized by reacting petroleum and carboxylic acids with allylic alcohol in the presence of an N-

methylpyrrolidone hydrogen sulfate- ionic liquid catalyst. As a result obtained ethers were subjected to hydrochlorination.

Based on the obtained chlorine-containing compound and the synthesized amidoamine and imidazoline of natural petroleum acids, chlorine complexes were obtained, which were tested as inhibitors in CO₂, H₂S and bio-corrosion. Moreover the physicochemical properties of chlorine-containing complexes of petroleum acids were studied. It was found that the synthesized complexes at concentrations of 5, 10 and 15 mg/l exhibit a bactericidal effect and completely suppress the growth of bacteria [4-8].

These complexes have relatively higher antimicrobial activity against sulfate-reducing bacteria than the industrial bactericide-inhibitors AMDOR-IK-7 and AMDOR-IK-10, taken as a standard. In addition, the synthesized complexes were tested as H₂S and CO₂ corrosion inhibitors, which at 100 ppm CO₂ corrosion protects metal surfaces from corrosion by 91-95%, and in the case of H₂S corrosion at 5, 10, 15 ppm, amidoamine (imidazoline) chloride complexes protect PNA 100% metal surface against H₂S corrosion.

The resulting complexes also have surface-active properties, since it is known that one of the methods of combating oil spills at sea is the use of surfactants, which, along with high oil-collecting and oil-dispersing ability, should be environmentally friendly. Tensiometric measurements revealed the high surface activity of the synthesized chlorine complexes at the water-air interface and calculated its colloidal chemical parameters. Conducted laboratory methods have shown that the reagent itself and its 5% aqueous solution have a high oil-collecting ability.



Surface tension isotherms of complexes

Surface tension isotherms of complexes 1- amidoamine dichlorodipropyl α -naphthylacetic complex of NOA, 2- amidoamine chloropropyl benzoate of NOA, 3- amidoamine allyl chlorine complex of NOA, 4- amidoamine dichloro dipropyl phthalate of NOA, 5- amidoamine dichloro dipropyl adipinate of NOA.

The synthesized complexes, diluted in distilled water, have a mixed oil-collecting-oil-dispersing effect. In distilled water, in the case of a 5% aqueous solution of the reagent, dispersion of the oil film is mainly observed. In fresh water, when using 100% and diluted reagent, a mixed effect of oil dispersion-oil collection-oil dispersion is determined. In seawater, the reagent in undiluted form is an oil dispersant, and diluted form it

shows a mixed effect of oil dispersing-oil gathering-oil dispersing properties. Activity of reagent is 3-4 days.

Specific electrical conductivities of imidazoline complexes of synthesized surfactants were determined by conductometric method. The results of the specific electrical conductivity of the synthesized complexes are presented in the table.

Specific electrical conductivity of aqueous solutions of complexes

Complexes	Reagent concentration in water, % (by mass)								
	0,001	0.0025	0.005	0.01	0.015	0.02	0.04	0.06	0.1
	Specific electrical conductivity, $\mu\text{S}/\text{sm}$ (at 20°C)								
Imidazolin dichlorodipropyl α -naphthylacetic complex of NOA (1)	2,4	3,8	6,0	14,2	17,0	21,8	47,6	68,3	99,3
Imidazolin chloropropyl benzoate of NOA (2)	3,6	5,2	8,7	19,1	23,1	28,9	61,3	91,0	130,5
TNT Imidazolin propilxlor(3)	3	4,1	6,5	14,4	17,3	22,1	43	63,8	91,65
Imidazoline dichlorodipropyl phthalate of NOA at (4)	3,2	4,8	7,8	17,8	21,7	26,8	52,4	76,5	111,7
Imidazoline dichloro dipropyl adipinate of NOA. (5)	5,0	7,6	13,1	32,4	39,7	51,6	101,5	147	210,6

As can be seen from the table, as the concentration of solutions increases, the values of specific electrical conductivity increase. The highest electrical conductivity in the range of 0.001-0.1% shows the imidazolin chloropropyl benzoate complex (3,6-130,5 $\mu\text{S}/\text{sm}$) and imidazolin dichlorodipropyl adipinate complex (5,0-210.6 $\mu\text{S}/\text{sm}$). The lowest electrical conductivity in the range of 0.001-0.1% shows the imidzaolin dichlorodipropyl α -naphthylacetic complex (2.4-99.3 $\mu\text{S}/\text{sm}$) and TNT imidazolin propilxlor complex (3-91,65 $\mu\text{S}/\text{sm}$).

References

1. Воробьев Ю.Л., Акимов В.А., Соколов Ю.И. Предупреждение и ликвидация аварийных разливов нефти нефтепродуктов. –М.: Ин-октаво, 2005.
2. Асадов З.Г., Саламова Н.В., Ахмедова Г.А., Зарбалиева И.А. Получение, физико-химические характеристики, нефтесобирающие и диспергирующие свойства оксипропиловых эфиров фракции

кислот рыбьего жира и их фосфатов // Естественные и технические науки. 2009, №3 (41), с. 64-68

3. Asadov Z.H., Nasibova Sh.M., Rahimov R.A., Ahmadova G.A., Huseynova S.M., Surface activity and micellization parameters of cationic surfactants containing hydroxyethyl group and C9-chain, Journal of Molecular Liquida, 2017, vol. 225, p. 451-455.

4. Mamedova N.A., Abbasov V. M., Mamedkhnova S.A, Axmadbayova S.F. Synthesis and research into anticorrosion and bactericide properties of chlorine-containing amidiamin complexes of distilled natural oil acids// Institute of Petrochemical Processes, Baku, 2018, pp180-190.

5. Mamedova N.A, Mamedkhnova S.A. Synthesis and research into anticorrosion and bactericide properties of chlorine-containing imidazoline complexes of distilled natural oil acids//Chemical problems, 2018, Baku № 3(16), pp 343-350

6. Mamedova N.A., Mamedkhanova S.A. Syn-

thesis and study of the bactericidal properties of chlorine-containing amidoamine complexes of natural petroleum acids. Eurasian Union of Scientists (ESU), No. 5(50), 2018, p. 66-72

7. Mamedova N. A., Mamedkhanova S. A., Shakhmamedova A. G. Synthesis and application of salts and complexes based on allyl ester of petroleum

acids as inhibitors of carbon dioxide corrosion. Petrochemistry, Moscow, 2019, volume 59, no. 2, With. 207–210

8. Mamedova N. A., Mamedkhanova S. A. Synthesis and study of bactericidal properties of chlorine-containing amidoamine complexes of natural petroleum acids/ Eurasian Union of Scientists– 2018. – № 5-2(50). – p. 66-72